Machine Learning-Guided Design of High-Entropy FeCoCrMnCu Layered Double Hydroxides for Efficient Oxygen Evolution in Alkaline Media

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Table S1. Initial 70 experimental dataset used for ML training.

Sample	Fe	Co	Cr	Mn	Cu	Overpotential
1	0.1	0.1	0.6	0.1	0.1	301
2	0.5	0.5	0	0	0	365
3	0	1	0	0	0	370
4	0	0	0	1	0	379
5	0.2	0	0.5	0.1	0.2	311
6	0.5	0	0.5	0	0	368
7	0.5	0	0	0	0.5	379
8	0.1	0.1	0.1	0.7	0	335
9	0.1	0.2	0.1	0.4	0.2	303
10	0.35	0.1	0.1	0.1	0.35	340
11	0	0	0.5	0.5	0	366
12	0.1	0.1	0.1	0.1	0.6	335
13	0	0.25	0.25	0.25	0.25	364
14	0	0	0	0.5	0.5	394
15	1	0	0	0	0	408
16	0.1	0.6	0.1	0.1	0.1	353.5
17	0	0.5	0	0	0.5	390
18	0	0	0.5	0	0.5	380
19	0.5	0	0	0.5	0	378
20	0	0	0	0	1	396
21	0.2	0.4	0.1	0	0.3	341
22	0.7	0.2	0.1	0	0	348
23	0	0	1	0	0	364
24	0	0.5	0	0.5	0	388
25	0	0.5	0.5	0	0	365
26	0.1	0.1	0	0.2	0.6	361
27	0	0.5	0.1	0.4	0	365
28	0.2	0	0.3	0.4	0.1	294
29	0.2	0.1	0.1	0.6	0	335
30	0.1	0.2	0.7	0	0	307
31	0.3	0.1	0.3	0	0.3	305
32	0.1	0.4	0.3	0.2	0	331
33	0.4	0.3	0	0.1	0.2	344
34	0.5	0.2	0.1	0.2	0	335
35	0	0	0.1	0.8	0.1	383
36	0	0.7	0.1	0.2	0	369
37	0	0.4	0.1	0	0.5	383
38	0.3	0.2	0.4	0	0.1	286
39	0.4	0.2	0.1	0	0.3	335
40	0.2	0.1	0.5	0.2	0	313
41	0.6	0.1	0	0	0.3	369
42	0.3	0.1	0	0.2	0.4	333
43	0.1	0.4	0.5	0	0	304

44	0	0	0.8	0	0.2	362
45	0.5	0	0.3	0.1	0.1	317
46	0	0	0.3	0.5	0.2	376
47	0.4	0	0.1	0.1	0.4	343
48	0.2	0	0.1	0.4	0.3	338
49	0.3	0.4	0.2	0	0.1	309
50	0	0.2	0.2	0.1	0.5	377
51	0	0.1	0.6	0.3	0	365
52	0.1	0.1	0	0.6	0.2	303
53	0	0	0.4	0.1	0.5	381
54	0	0.2	0.5	0.1	0.2	364
55	0.4	0	0.1	0.4	0.1	321
56	0	0.4	0	0.2	0.4	387
57	0.2	0.3	0.1	0.4	0	332
58	0.2	0.7	0	0	0.1	338
59	0	0	0.1	0.4	0.5	390
60	0	0.3	0.1	0.6	0	356
61	0.1	0.1	0.1	0.1	0.6	311
62	0.1	0.1	0.1	0.6	0.1	324
63	0.1	0.6	0.1	0.1	0.1	355
64	0.6	0.1	0.1	0.1	0.1	349
65	0.1	0.35	0.35	0.1	0.1	312
66	0.1	0.35	0.1	0.1	0.35	367
67	0.1	0.1	0.1	0.35	0.35	345
68	0.1	0.35	0.1	0.35	0.1	357
69	0.35	0.1	0.1	0.35	0.1	340
70	0.35	0.35	0.1	0.1	0.1	353

Table S2. Additional data sets for ML-guided FeCoCrMnCu-LDHs compositions.

Sample	Fe	Co	Cr	Mn	Cu	Predicted η	Experimental η	Error (%)
1	0.20	0.20	0.20	0.20	0.20	296.0283	307.07	3.60
2	0.50	0	0.10	0.15	0.25	334.7695	345.01	2.97
3	0.75	0.15	0.05	0.05	0	353.2497	365.20	3.27

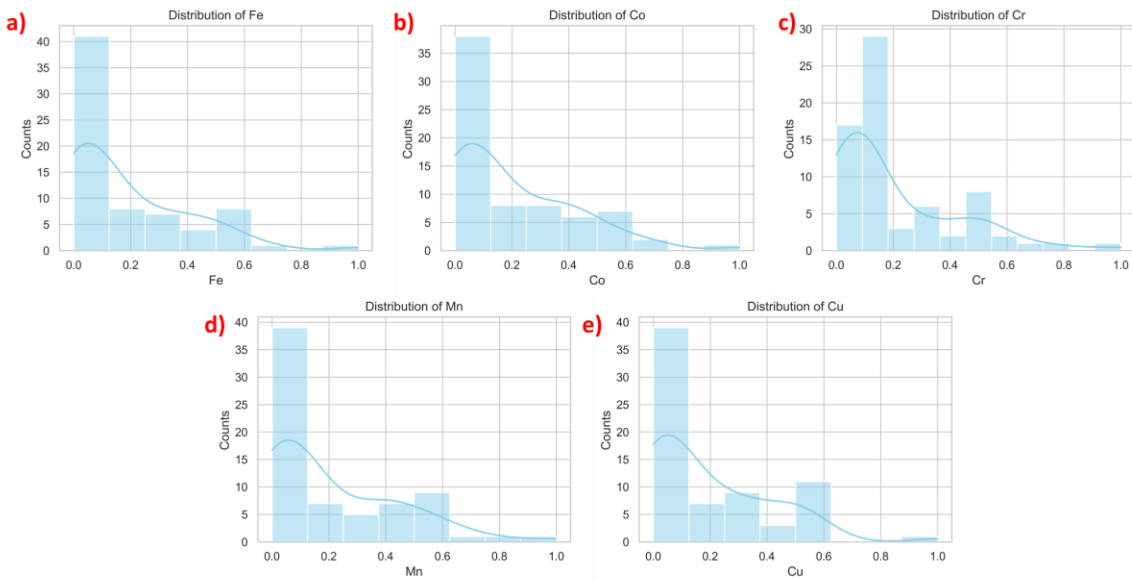

Fig. S1. Histogram of elemental distribution in the initial dataset: (a) Fe, (b) Co, (c) Cr, (d) Mn, and (e) Cu.

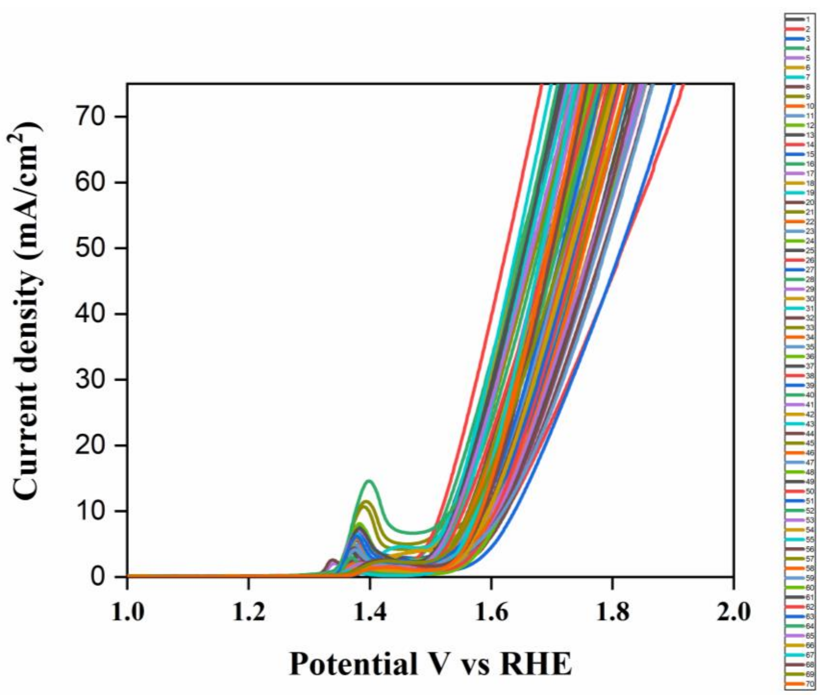

Fig. S2. LSV curves of 70 different high-entropy FeCoCrMnCu LDHs.

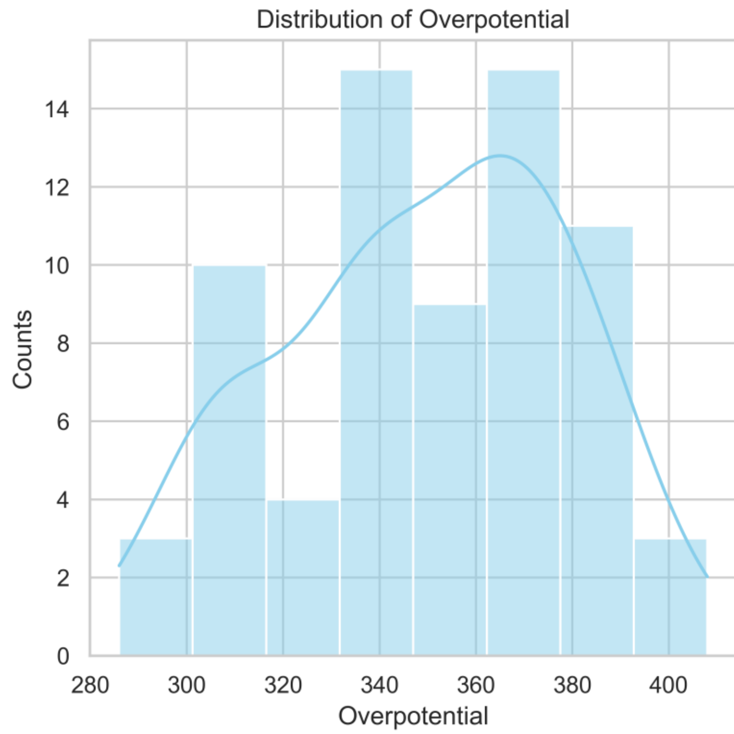

Fig. S3. Histogram of overpotential distribution in the initial dataset.

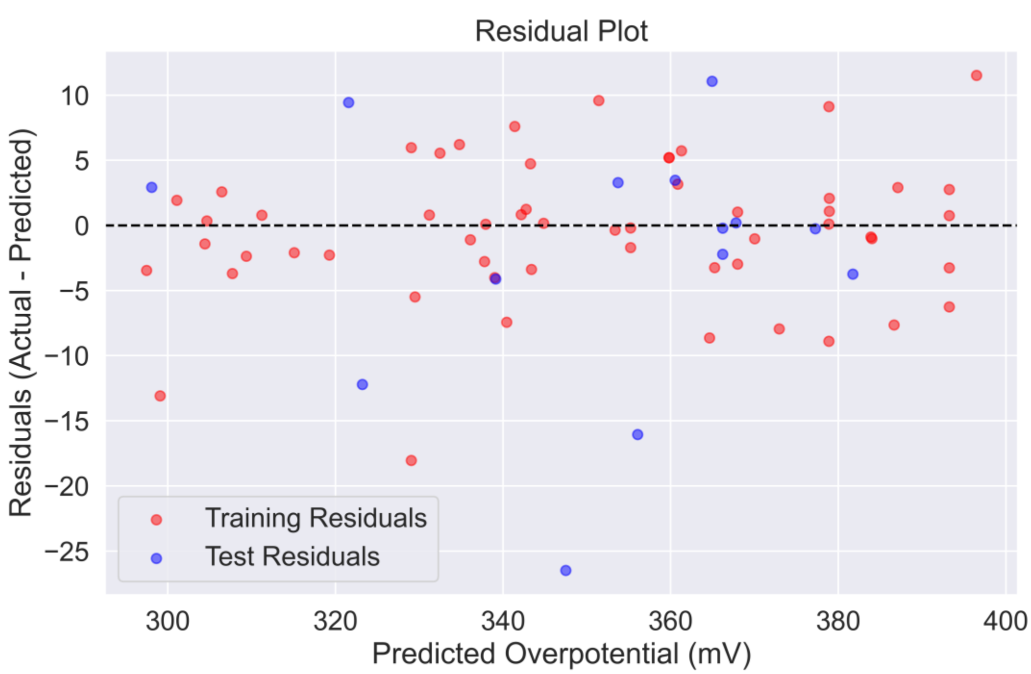

Fig. S4. Residual plot.

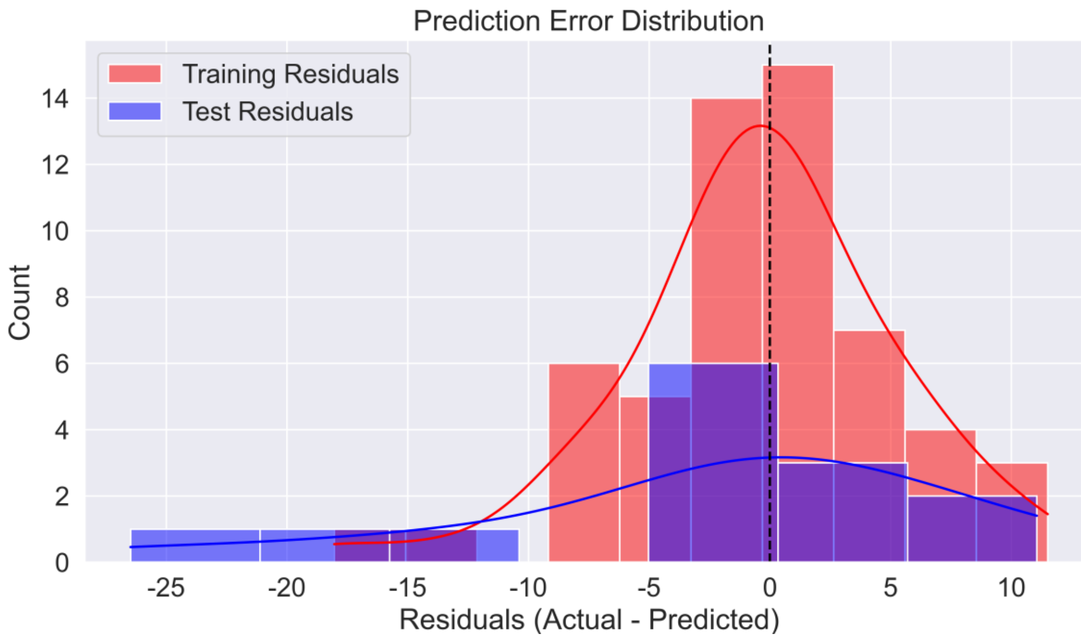

Fig. S5. Distribution plot of prediction errors.

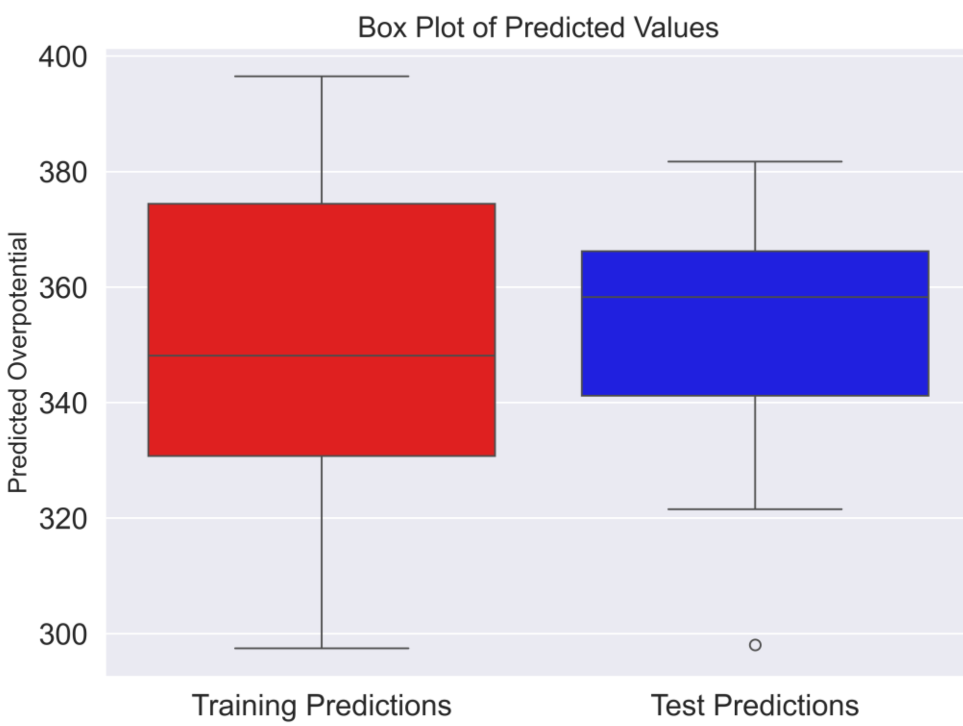

Fig. S6. Box plot of predicted values.

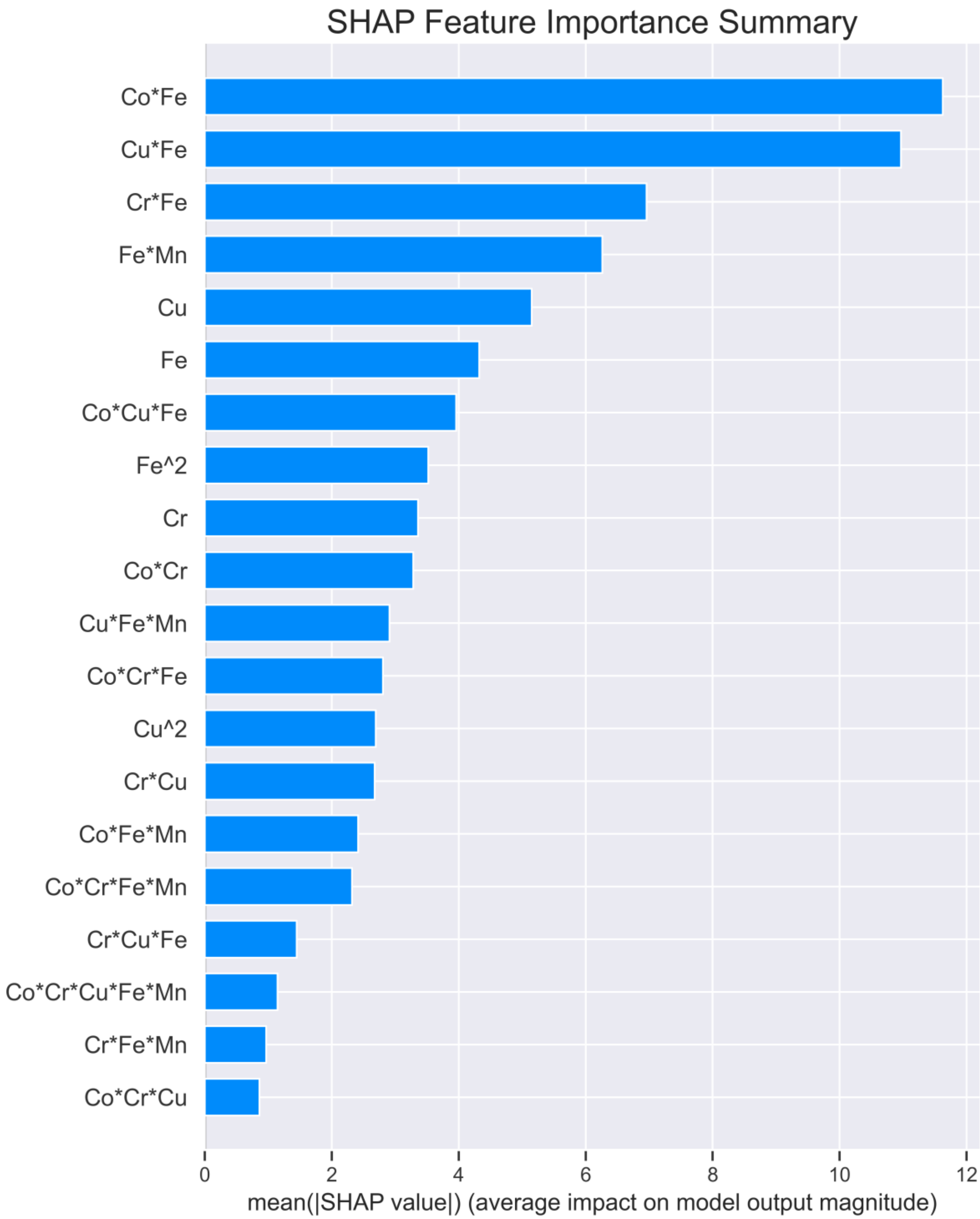

Fig. S7. Mean absolute SHAP values.

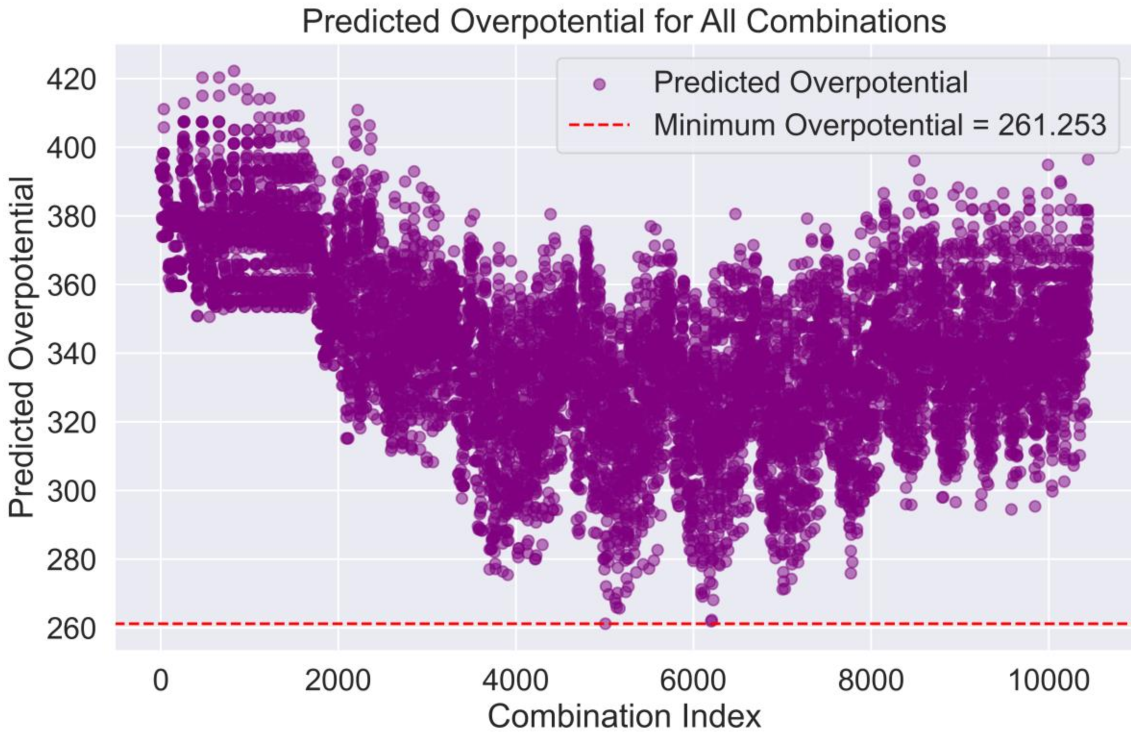

Fig. S8. Distribution plot of predicted overpotential.

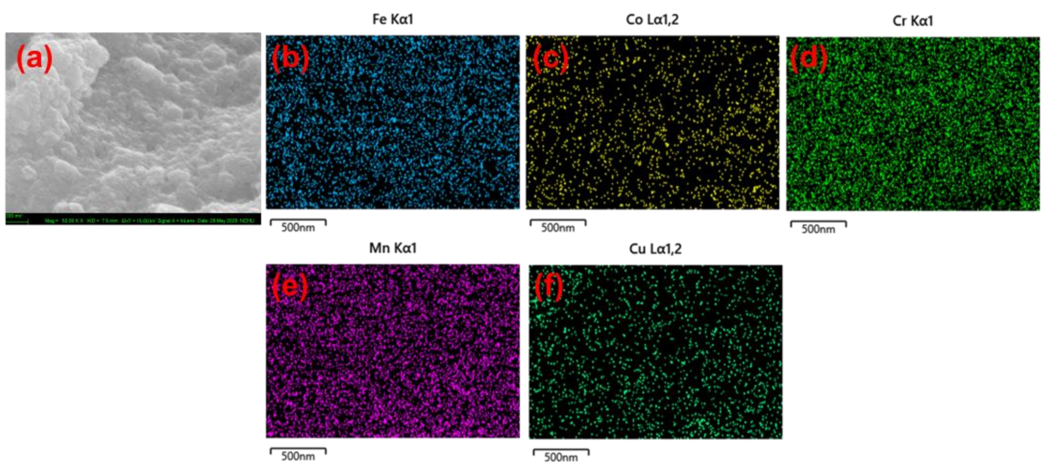

Fig. S9. (a) Bright field image and (b-f) EDX mappings of Fe, Co, Cr, Mn, and Cu elements of optimized FeCoCrMnCu LDHs.

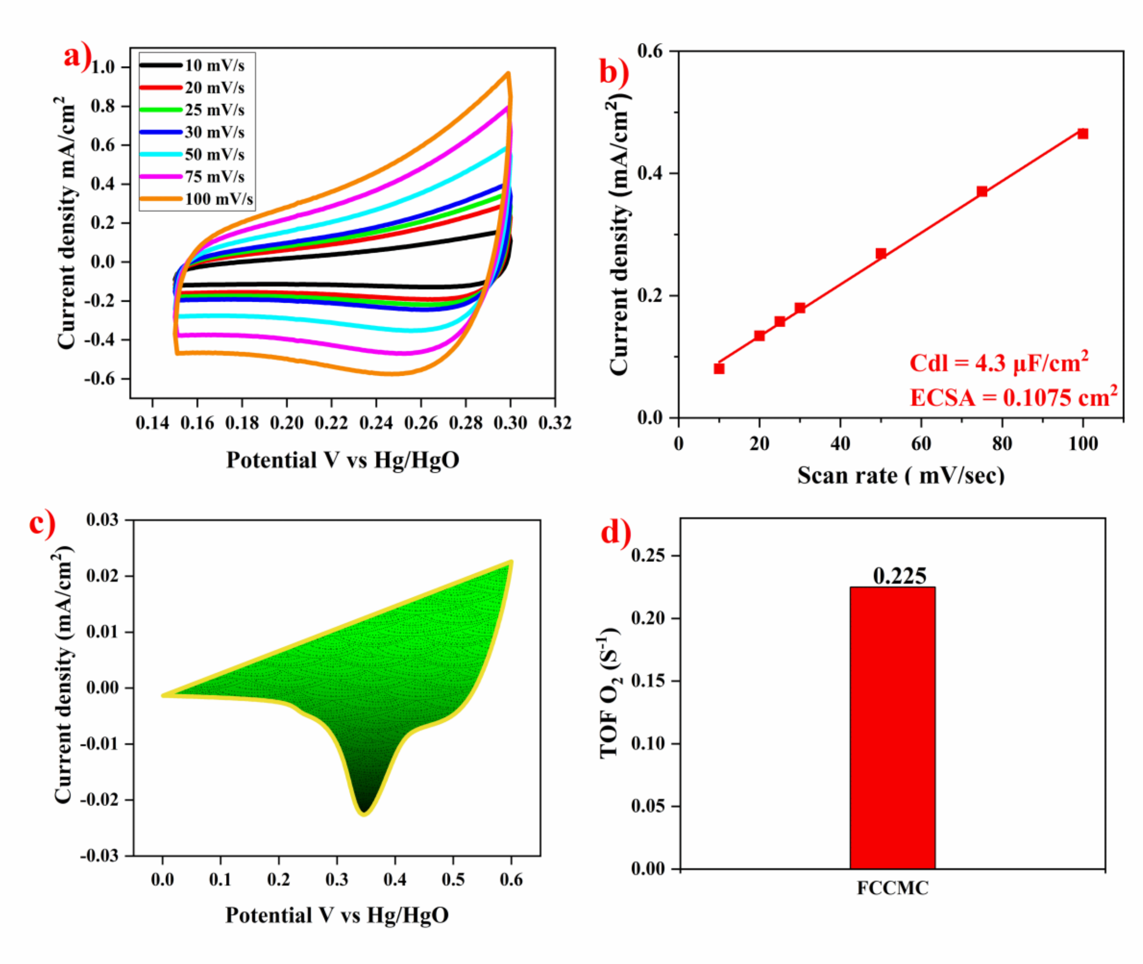

Fig. S10. Electrochemical characterization of the optimized FeCoCrMnCu LDHs includes: (a) CV curves at various scan rates, (b) linear fitting of current density versus scan rate, (c) reduction peak area, and (d) TOF at 400 mV.

Link for experimental dataset

https://github.com/NCHU-UPICE/Dataset-of-the-compositional-variations-in-high-entropy-FeCoCrMnCu-layered-double-hydroxides-LDHs-.git